

BOOK ANALYTICS USING DATA SCIENCE

End-to-End Data Science Capstone Project

Prepared By:Ritesh Redekar

Domain:Book Analytics

Dataset Source:Open Library API

Tools Used:

Python,Pandas,NumPy,Matplotlib,Scikit-Learn,Power BI

Project Objective:

- **Objective**
- The objective of this project is to collect real-world book data from the Open Library API and transform it into meaningful business insights using Data Science techniques.
- **This project includes:**
- Web Scraping
Data Cleaning
Feature Engineering
Exploratory Data Analysis
Machine Learning
Classification
Interactive Power BI Dashboard

Workflow:

Project WorkflowOpen Library API



Web Scraping



Raw Dataset



Data Cleaning



Feature Engineering



Exploratory Data Analysis



Machine Learning



Power BI Dashboard

Web Scrapping:

- **Data Collection**
- Book data was collected using the Open Library REST API.
- Python libraries used:
- Requests
- Pandas
- **Collected information:**
- Title
- Author
- Subject
- Publish Year
- Edition Count
- Work ID
- **Total Records Collected**
- 1877 Books

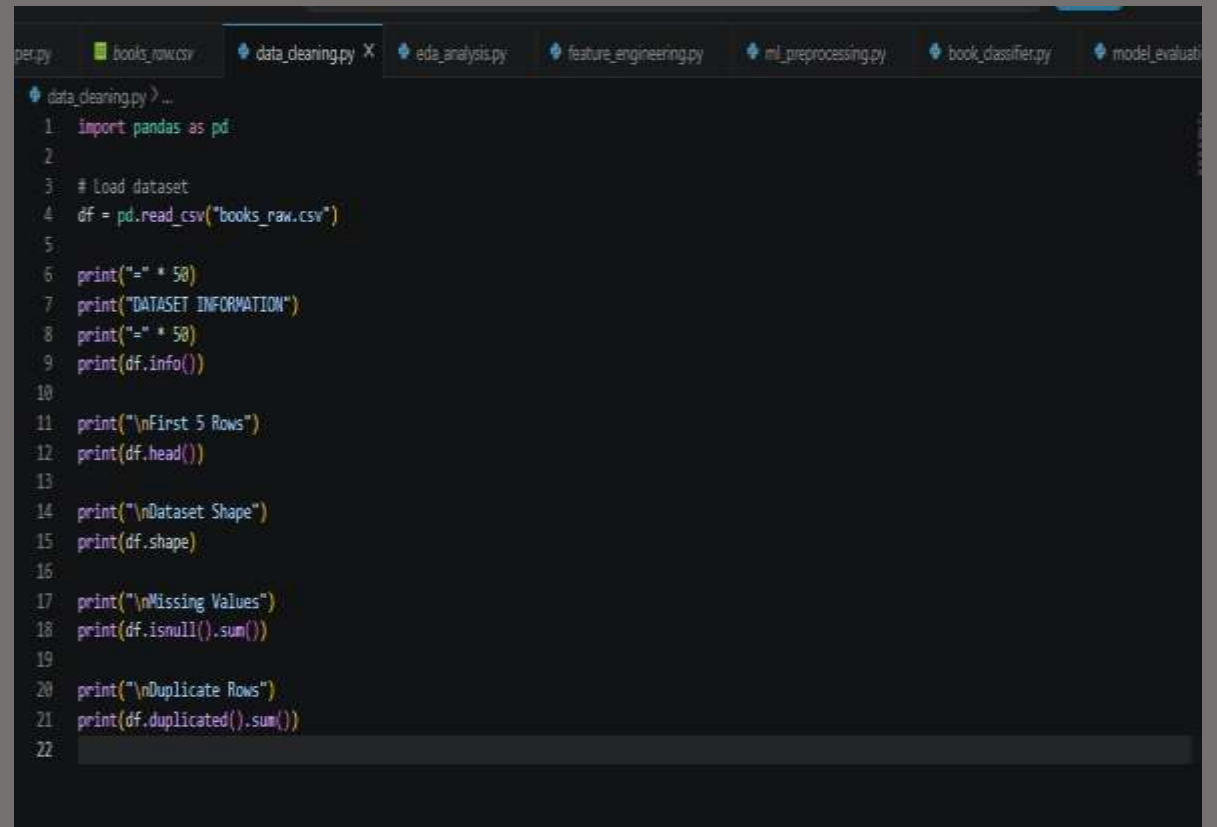
```
books_scraper.py  data_cleaning.py  data_analysis.py  feature_engineering.py  ml_preprocessing.py  book_classifier.py  model_evaluation.py
books_scraper.py
1 import requests
2 import pandas as pd
3
4 # List of subjects to collect books from
5 subjects = [
6     "fantasy",
7     "science_fiction",
8     "romance",
9     "history",
10    "mystery",
11    "adventure",
12    "horror",
13    "children",
14    "biography",
15    "love",
16    "thriller",
17    "poetry",
18    "drama",
19    "comics",
20    "fiction",
21    "art",
22    "music",
23    "travel",
24    "science",
25    "technology",
26    "education",
27    "business",
28 ]
```

```
books_data.py
1 Title,Author,Publish Year,Edition Count,Subject,Work ID
2 Alice's Adventures in Wonderland,Lewis Carroll,1865,3547,Fantasy,/works/OL138052W
3 The Wonderful Wizard of Oz,L. Frank Baum,1899,2049,Fantasy,/works/OL18417W
4 Treasure Island,Robert Louis Stevenson,1880,1908,Fantasy,/works/OL2003W
5 Gulliver's Travels,Jonathan Swift,1726,1810,Fantasy,/works/OL1000W
6 A Midsummer Night's Dream,William Shakespeare,1600,1501,Fantasy,/works/OL25061W
7 The Prince,Niccolò Machiavelli,1515,1487,Fantasy,/works/OL18807W
8 Through the Looking-Glass,Lewis Carroll,1865,1379,Fantasy,/works/OL15140W
9 The Wind in the Willows,Kenneth Grahame,1908,1146,Fantasy,/works/OL20570W
10 Five Children and It,Edith Nesbit,1903,720,Fantasy,/works/OL4549W
11 The Princess and the Goblin,George MacDonald,1872,740,Fantasy,/works/OL15440W
12 The Lost World,Arthur Conan Doyle,1900,767,Fantasy,/works/OL16180W
13 The Marvelous Land of Oz,L. Frank Baum,1900,724,Fantasy,/works/OL18390W
14 Ome of Oz,L. Frank Baum,1907,701,Fantasy,/works/OL1841W
15 Le petit prince,Antoine de Saint-Exupéry,1943,680,Fantasy,/works/OL1006W
16 The Emerald City of Oz,L. Frank Baum,1910,674,Fantasy,/works/OL18400W
17 Dorothy and the Wizard in Oz,L. Frank Baum,1908,622,Fantasy,/works/OL18412W
18 The Last Princess of Oz,L. Frank Baum,1917,591,Fantasy,/works/OL18413W
19 The Story of the Amulet,Edith Nesbit,1900,578,Fantasy,/works/OL40747W
20 The Complete Life and Adventures of Santa Claus,L. Frank Baum,1902,569,Fantasy,/works/OL26235W
21 Alice's Adventures in Wonderland / Through the Looking Glass,Lewis Carroll,1865,544,Fantasy,/works/OL13141W
22 Winston Eighty-Four,George Orwell,1940,536,Fantasy,/works/OL11688W
23 The Road to Oz,L. Frank Baum,1909,535,Fantasy,/works/OL26239W
24 The Sea Fairies,L. Frank Baum,1915,528,Fantasy,/works/OL26238W
25 Baum's American Fairy Tales,L. Frank Baum,1905,514,Fantasy,/works/OL26234W
26 Sky Island,L. Frank Baum,1912,510,Fantasy,/works/OL26230W
27 The Scarecrow of Oz,L. Frank Baum,1913,503,Fantasy,/works/OL26232W
```

Data Cleaning & Feature Engineering:

- Data Cleaning
 - **Performed:**
 - Removed missing values
 - Removed duplicates
 - Converted data types
 - Standardized text
 - Created new features
-
- **New Features Created**
 - Book Age
 - Publication Decade
 - Edition Category

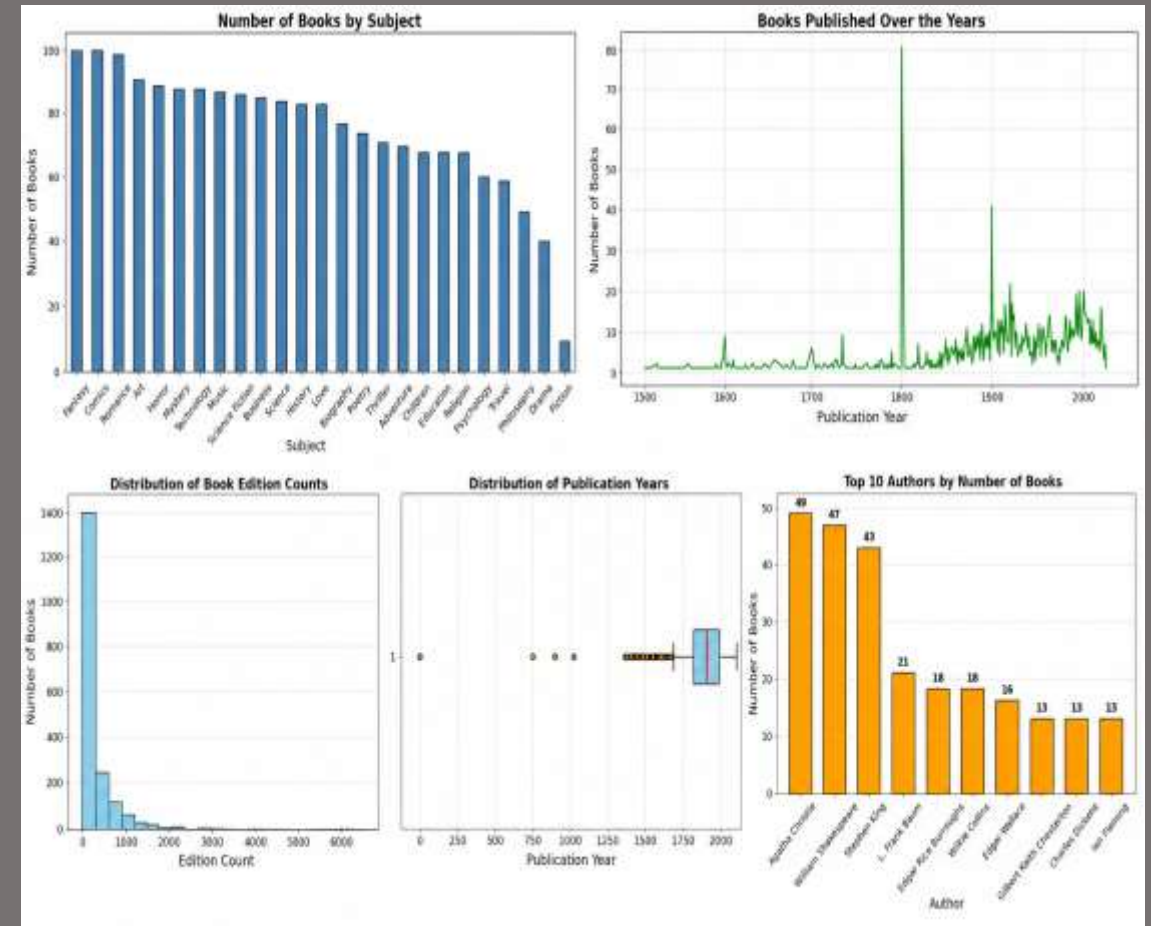
Data_cleaning.py



```
data_cleaning.py > ...
1. import pandas as pd
2.
3. # Load dataset
4. df = pd.read_csv("books_raw.csv")
5.
6. print("=" * 50)
7. print("DATASET INFORMATION")
8. print("=" * 50)
9. print(df.info())
10.
11. print("\nFirst 5 Rows")
12. print(df.head())
13.
14. print("\nDataset Shape")
15. print(df.shape)
16.
17. print("\nMissing Values")
18. print(df.isnull().sum())
19.
20. print("\nDuplicate Rows")
21. print(df.duplicated().sum())
22.
```

Exploratory Data Analysis:

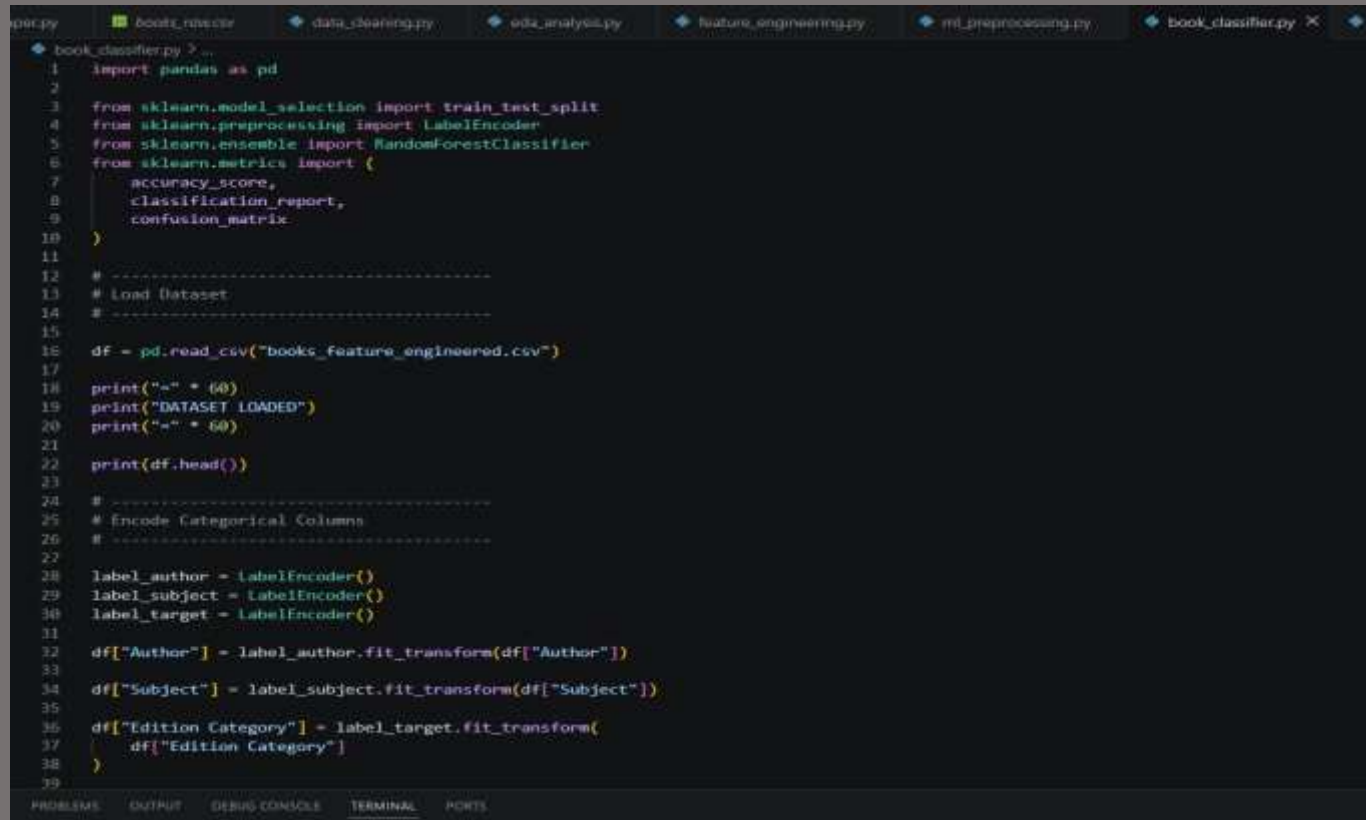
- EDA
- Several visualizations were created to identify trends.
- **Visualizations:**
- Books by Subject
- Books by Publication Year
- Publication Year Box
- plotTop Authors
- Edition Count Distribution
- **Key Findings**
- Romance and Fantasy are highly represented.
- Most books were published after 1950.
- Few authors contribute many books.
- Edition Count varies significantly.



Machine Learning:

- **Model Building**
- Problem Type
- Classification
- Target Variable
- Edition Category
- Algorithm Used
- Random Forest Classifier
- Train-Test Split
- 80 : 20
- Libraries
- Scikit-Learn

Book_classifier.py



```
1 import pandas as pd
2
3 from sklearn.model_selection import train_test_split
4 from sklearn.preprocessing import LabelEncoder
5 from sklearn.ensemble import RandomForestClassifier
6 from sklearn.metrics import (
7     accuracy_score,
8     classification_report,
9     confusion_matrix
10 )
11
12 # -----
13 # Load Dataset
14 # -----
15
16 df = pd.read_csv("books_feature_engineered.csv")
17
18 print("-" * 60)
19 print("DATASET LOADED")
20 print("-" * 60)
21
22 print(df.head())
23
24 # -----
25 # Encode Categorical Columns
26 # -----
27
28 label_author = LabelEncoder()
29 label_subject = LabelEncoder()
30 label_target = LabelEncoder()
31
32 df["Author"] = label_author.fit_transform(df["Author"])
33
34 df["Subject"] = label_subject.fit_transform(df["Subject"])
35
36 df["Edition Category"] = label_target.fit_transform(
37     df["Edition Category"]
38 )
39
```

Model Evaluation:

- **Model Performance**
- Training Samples:
- 1501
- Testing Samples:
- 376
- Accuracy:
- **98.94%**
- **Classification Metrics**
- Precision
- Recall
- F1 Score
- Confusion Matrix Generated

Terminal_output:

```
book_classifier.py
124 print(confusion_matrix(y_test, y_pred))

PS C:\Users\rites\OneDrive\Desktop\capstone_project> python book_classifier.py
DATASET LOADED
=====
Title Author Publish Year ... Book Age Publication Decade Edition Category
0 Alice's Adventures in Wonderland Lewis Carroll 1865 ... 160 1860 High
1 The Wonderful Wizard of Oz L. Frank Baum 1899 ... 126 1890 High
2 Treasure Island Robert Louis Stevenson 1888 ... 145 1880 High
3 Gulliver's Travels Jonathan Swift 1726 ... 299 1720 High
4 A Midsummer Night's Dream William Shakespeare 1600 ... 425 1600 High
[5 rows x 9 columns]
Feature Shape : (1877, 6)
Target Shape : (1877,)
Training Samples : 1501
Testing Samples : 376

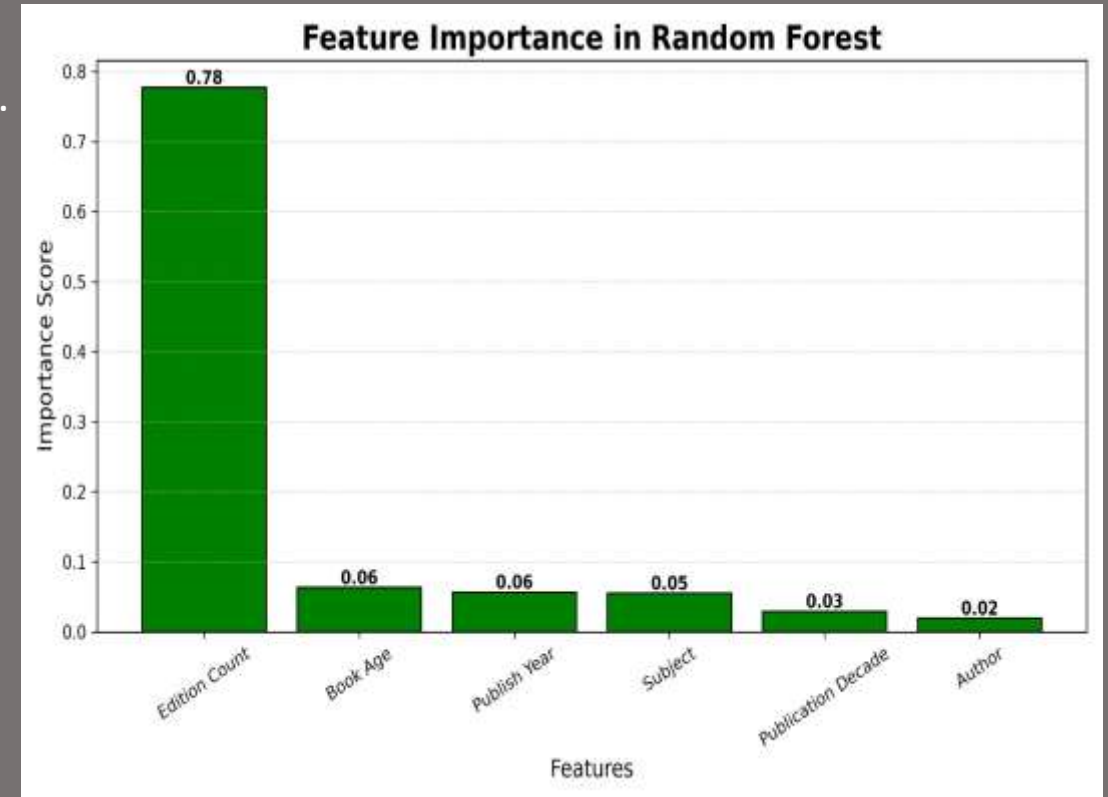
=====
MODEL ACCURACY
=====
Accuracy : 98.94%

Classification Report
=====
precision recall f1-score support
High 1.00 0.90 0.95 39
Low 1.00 1.00 1.00 195
Medium 0.97 1.00 0.99 142
accuracy 0.99
macro avg 0.99 0.97 0.98 376
weighted avg 0.99 0.99 0.99 376

Confusion Matrix
[[ 35  0  4]
 [ 0 195  0]
 [ 0  0 142]]
PS C:\Users\rites\OneDrive\Desktop\capstone_project>
```

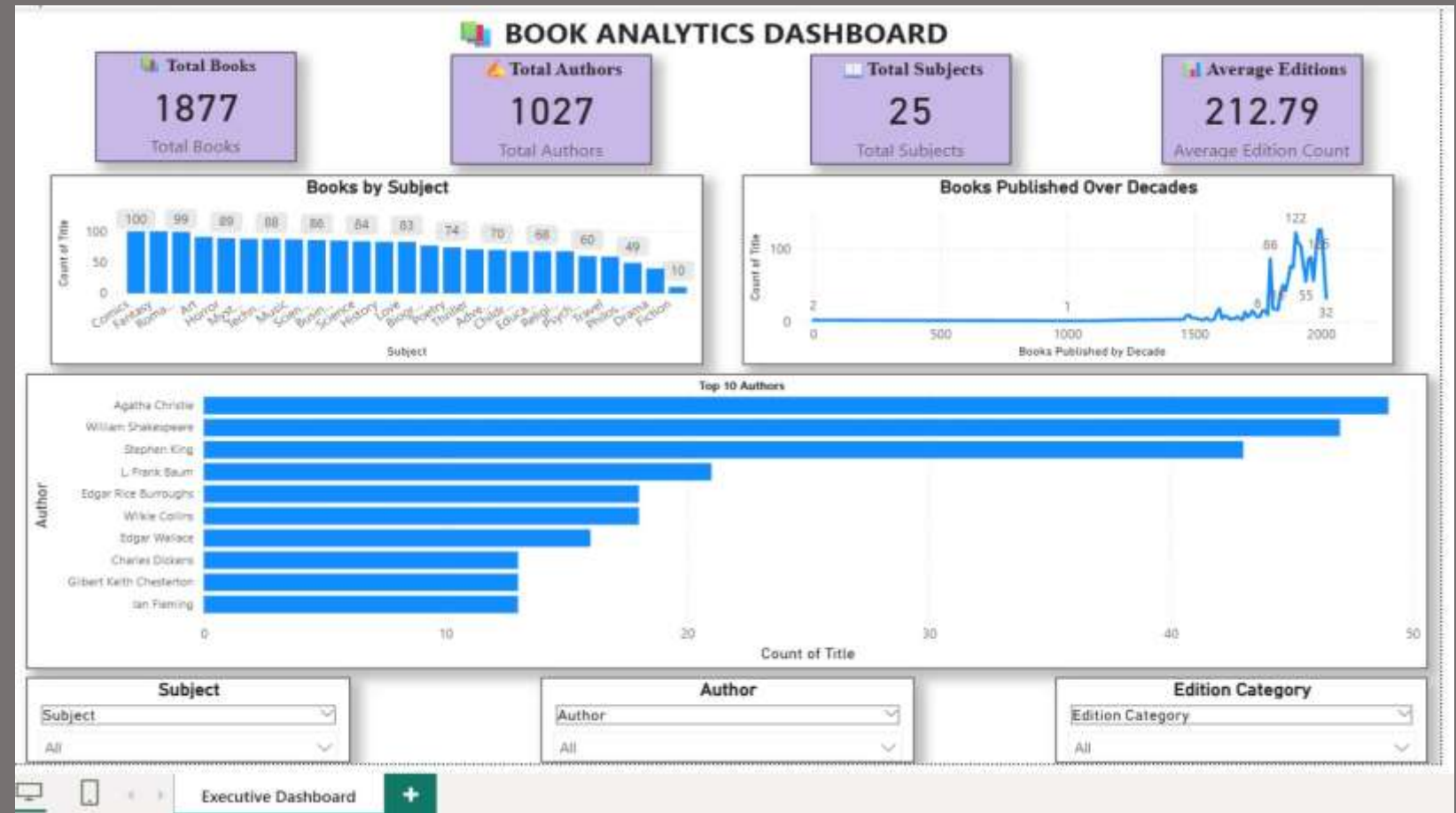

Feature Importance:

- **Feature Importance**
- Random Forest identified the most important features.
- Ranking
- 1 Edition Count
- 2 Book Age
- 3 Publish Year
- 4 Subject
- 5 Publication Decade
- 6 Author
- Observation
- Edition Count contributes nearly 78% towards prediction



Power BI Dashboard:

- Interactive Dashboard
- Dashboard contains
- KPIs
- Total Books
- Total Authors
- Total Subjects
- Average Editions
- Charts
- Books by Subject
- Books Published Over Decades
- Top 10 Authors
- Interactive Filters
- Subject
- Author
- Edition Category



Business Insights:

- **Key Insights**
- **1877** books were analyzed.
- More than 1000 unique authors exist.
- Romance and Fantasy contain the highest number of books.
- Book publication increased significantly after the 1950s.
- Edition Count is the strongest predictor of Edition Category.
- The Random Forest model achieved 98.94% accuracy

Conclusion:

- **Conclusion**
- The project successfully demonstrates an end-to-end Data Science workflow.
- Achievements
 - ✓ ☐ **Web Scraping**
 - ✓ ☐ **Data Cleaning**
 - ✓ ☐ **Feature Engineering**
 - ✓ ☐ **Exploratory Data Analysis**
 - ✓ ☐ **Machine Learning**
 - ✓ ☐ **Interactive Dashboard**
- The generated insights can help publishers and libraries understand publishing trends and book popularity.

Thank You & Video Submission:

- **THANK YOU**
- **Prepared By:Ritesh Redekar**
- Project Demonstration Video

Video url: <https://drive.google.com/file/d/1tec7h-P2rY07690xwnBXSjbrRtB5u1/view?usp=sharing>